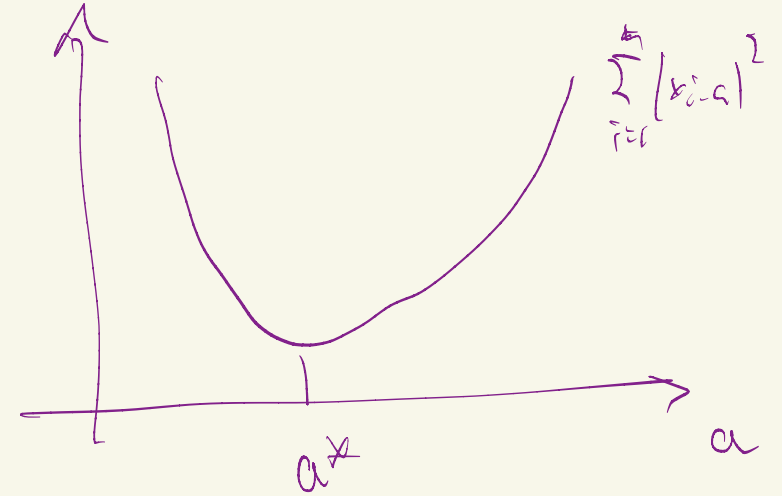


Eco 202 - Fall 2026

Class 02

Problem:

$$\min_a \sum_{i=1}^n (x_i - a)^2$$



FOC:

$$\frac{\partial}{\partial a} \sum_{i=1}^n (x_i - a)^2 = \sum_{i=1}^n \frac{\partial}{\partial a} (x_i - a)^2$$

$$= \sum_{i=1}^n 2(x_i - a)(-1) = 0$$

$$\Leftrightarrow \sum_{i=1}^n (x_i - a) = 0$$

$$\Leftrightarrow \sum_{i=1}^n (x_i - a) = 0$$

$$\Leftrightarrow \sum_{i=1}^n x_i - \sum_{i=1}^n a = 0$$

$$\Leftrightarrow \sum_{i=1}^n x_i - n \cdot a = 0$$

$$\Leftrightarrow \sum_{i=1}^n x_i = n \cdot a$$

$\Rightarrow$

$$a^* = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\sum_{i=1}^n y_i = y_1 + y_2 + \dots + y_n$$

$$\sum_{i=1}^n a = \underbrace{a + a + \dots + a}_{n \text{ times}}$$